

Glossary of Terms: Bimodal Manifold Interaction (BMI)

I. Fundamental Architecture

Bimodal Manifold Interaction (BMI)

The theoretical framework positing that the universe is a 6D bulk manifold created by the collision of two membranes (branes σ_A and σ_B). All physical phenomena are derivations of the geometric relaxation of this collision.

Thick Brane Interface (Σ)

The 4D spacetime hypersurface of finite thickness δ where our observable reality exists. It acts as the "crease" or "boundary" where the two colliding branes interact.

Bridge Stress (τ)

The elastic tension and hyperspace momentum of the connection between the colliding branes. It is the primary component of gravitational field energy.

Inter-brane Potential (V_{gap})

The vacuum energy inherent to the spatial gap separating the colliding branes. It acts as the driver for global cosmological expansion (Dark Energy).

Metric Equilibrium (Δ)

The total net spacetime curvature of the manifold, defined as the sum of Bridge Stress and Inter-brane Potential ($\Delta = \tau + V_{\text{gap}}$).

II. Particle & Topological Mechanics

Harmonic Node

A discrete, stable configuration of localized energy within the interface. These are the BMI framework's equivalent to "particles."

Nested Torus (T^3)

The geometric topology of a harmonic node. Matter is not a point-particle but a stable, multi-layered phase-lock of toroidal membranes within the interface.

Winding Matrix (R_{top})

A mathematical operator that maps how complex nested geometries (e.g., meta-stable leptons) relax into lower-energy configurations, governing phenomena like radioactive decay.

Manifold Impedance Matrix (Z_{ij})

A complex tensor representing the background elasticity of the manifold. It determines the strength and range of forces between harmonic nodes.

Topological Pinning

The causal, mechanical process wherein an expanding wave function localizes into a singular spatial coordinate due to local manifold curvature.

Topological Breaking Limit (Ω_{max})

The absolute energy threshold where local energy density exceeds the elastic modulus of the background vacuum. Beyond this, stable nested torus geometry becomes impossible.

III. Forces & Gauge Interactions

Screening Gate Function (g_{eff})

A dynamic, non-linear function that modulates interface permeability. It determines whether a field propagates as a diffuse wave or "pins" to a target based on local Ricci curvature.

Hyperspatial Shear Tensor ($S_{\mu\nu}$)

The tensor defining the structural alignment of the interface crease relative to the compactified shadow manifold. It dictates the directionality and momentum of particle paths.

Geometric Clamping

The mechanism of mass generation. Mass arises from the geometric resistance encountered when a gauge field attempts to twist perpendicular to the interface boundaries.

Shadow Manifold (M_{shadow})

The compactified, non-observable region of the 6D bulk that anchors the trifold micro-junctions of strong-force-interacting particles.

IV. Methodology

Atomic File Overwrite

The rigorous protocol of replacing entire file trees during development rather than fragmented edits, preventing "State Drift" in human-AI collaboration.

State Drift

The tendency for complex systems to diverge from their original physical definitions over long periods of iteration.